June 2004

INTERNATIONAL GCSE

MARK SCHEME
MAXIMUM MARK: 80
SYLLABUS/COMPONENT: 0620/02 **CHEMISTRY**

ROOT 19

1	(a)		B, C, F (all needed); Only contain one type of atom NOT: contain one kind of molecule NOT: cannot be split using chemical means	[1] [1]
	(b)		C	[1]
	(c)	(i)	B	[1]
		(ii)	any gas with diatomic molecules e.g. chlorine, hydrogen, hydrogen chloride	[1]
	(d)	(i)	F	[1]
		(ii)	pencil 'leads'/in pencils/lubricant/in electrical conductors/for electrodes/ in tennis racquets/in golf clubs/hockey sticks etc	[1]
	(e)	(i)	substance containing 2 or more different atoms combined/bonded/joined (both parts needed for mark) ALLOW: elements (chemically) combined	[1]
		(ii)	methane	[1]
	(f)	(i)	8 electrons round chlorine and bonded pair with dot and cross = 2 ALLOW: all dots or all crosses Correct number of electrons but bonded pair not clearly on overlap = 1 NOT: molecules other than hydrogen chloride	[2]
		(ii)	covalent	[1]
		(iii)	blue litmus; (litmus) turns red	[1] [1]
		(iv)	pH2	[1]
		(v)	2	[1]
		(vi)	magnesium chloride NOT: formula	[1]
				Total = 17
2	(a)		insoluble particles/solids/dirt trapped/caught on stones; NOT: filter reacts with insoluble impurities NOT: impurities unqualified Water passes through/filtered OWTTE	[1] [1]
	(b)	(i)	kill bacteria/germs, disinfect water OWTTE	[1]
		(ii)	neutralises acidity/water ALLOW: reacts with acids in water	[1]
		(iii)	calcium hydroxide NOT: formula	[1]
		(iv)	neutralising acid soils/neutralising acidic (industrial) waste/making bleaching powder/removing acidic gases/in Solvay process/in recovery of ammonia/making limewater/in water softening/for making plaster/for making mortar/controlling soil acidity NOT: neutralising acids unqualified NOT: making cement	[1]

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(c) **(i)** 100; [1]
°C (conditional on 100) [1]

(ii) anhydrous cobalt chloride/anhydrous copper sulphate (or correct colours); [1]
NOT: cobalt chloride/copper sulphate unqualified
Turns pink/blue (respectively) [1]

(iii) any suitable e.g. washing/cleaning/drinking/cooking [1]

(d) B [1]

(e) ethanol [1]
NOT: alcohol

(f) potassium hydroxide; hydrogen [1]
NOT: symbols

Total = 15

3 **(a)** means of measuring gas volume e.g. gas syringe/measuring cylinder (must be graduated); [1]
flask/test tube/vessel with calcium carbonate + acid leading to syringe etc
IGNORE: lack of reference to closed system (unless drawing incorrect) [1]
record volume on gas syringe/measuring cylinder/measure how much carbon dioxide given off [1]
at various time intervals/at a particular time [1]
OR
flask/vessel with calcium carbonate and hydrochloric acid in flask (1)
measure its mass at beginning of experiment (1)
measure mass of flask and contents during reaction (1)
at specific time(s) (1)

(b) **(i)** faster/greater/speeds up [1]

(ii) slower/less [1]

(iii) faster/greater/speeds up [1]

(c) **(i)** add aqueous sodium hydroxide; [1]
white precipitate; [1]
insoluble in excess [1]
(incorrect reagent = 0)
ALLOW: flame test - brick red

(d) **(i)** high in the reactivity series/very reactive [1]

(ii) 2 electrons in outer shell; [1]
inner shells correct as 2.8.8 [1]

Total = 13

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4	**(a)**		ethanol - solvent ethene - polymer bitumen - roads	[3]
	(b)		ethanol	[1]
	(c)	**(i)**	C	[1]
		(ii)	A	[1]
		(iii)	B	[1]
		(iv)	D	[1]
	(d)	**(i)**	(compound) containing <u>only</u> carbon and hydrogen NOT: it contains carbon and hydrogen	[1]
		(ii)	has only single bonds/ has general formula C_nH_{2n+2} NOT: it is saturated	[1]
				Total = 10
5	**(a)**		chlorine, argon, potassium, bromine, iodine ALLOW: symbols	[1]
	(b)		chlorine, potassium, argon, bromine, iodine ALLOW: symbols	[1]
	(c)		2nd box down ticked	[1]
	(d)		chlorine, bromine, iodine (all 3 needed) ALLOW: symbols	[1]
	(e)	**(i)**	potassium/K	[1]
		(ii)	argon/Ar	[1]
	(f)		1st and 4th boxes ticked (1 mark each)	[2]
	(g)	**(i)**	high (boiling point)	[1]
		(ii)	conducts/is high	[1]
	(h)		potassium loses <u>an</u>/<u>one</u> electron/loses outer shell chlorine gains <u>an</u>/<u>one</u> electron/outer shell becomes complete ALLOW: (for 1 mark) potassium loses two electrons + chlorine gains two electrons ALLOW: e.g. 2.8.8.1 → 2.8.8 for first mark Any indication of sharing electrons = 0	[1] [1]
				Total = 12

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6	**(a)**		carbon monoxide	[1]
	(b)		iron oxide loses oxygen/it loses oxygen/oxidation number of iron decreases ALLOW: iron gains electrons Answer must refer to the iron/iron oxide - therefore: NOT: carbon monoxide gains oxygen NOT: oxygen lost in the reaction NOT: iron loses oxygen	[1]
	(c)		3; 2 (one mark each)	[2]
	(d)	**(i)**	oxidise the impurities/oxidise Si or P or C/burn off the impurities NOT: get rid of impurities NOT: slag formation	[1]
		(ii)	exothermic	[1]
		(iii)	is/floats above the molten iron	[1]
		(iv)	calcium oxide	[1]
		(v)	stronger/harder/not brittle/less easily corroded ORA e.g. iron rusts NOT: less corrosive	[1]
	(e)		any 3 of: high melting/boiling points; have coloured compounds (NOT: they are coloured); have high densities; form complex ions; elements/compounds are good catalysts; form ions with different charges/variable oxidation states	[3]
	(f)		alloys	[1]

Total = 13

Grand Total = 80

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